LECTURE NOTES

Lecture 8 - Loss Functions cont.

AI506 — Advanced Machine Learning

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Information Theory

Shannon's Information theory (1948)

Information should have the following desirable properties

- Continuity
- Symmetry
- Maximal value
- Additive

Surprisal

Given some distribution $P(X)$, how can we caracterize how surprised we are when we observe a particular event. Intution: the smaller the probability of the event, the larger the surprise.

$$\text{Surprisal}(x) = \frac{1}{p(x)}$$

Then for multiple events

$$\text{Surprisal}(x, y) = \frac{1}{p(x)} \cdot \frac{1}{p(y)}$$

Then for additive property

$$h(x) = \log\left(\frac{1}{p(x)}\right) = -\log(p(x))$$

Entropy

Lets average Shannon Information

$$H(x) = \mathbb{E}[h(x)] = -\sum p(x)\log(p(x)),$$

also called "uncertainty of a random variable."

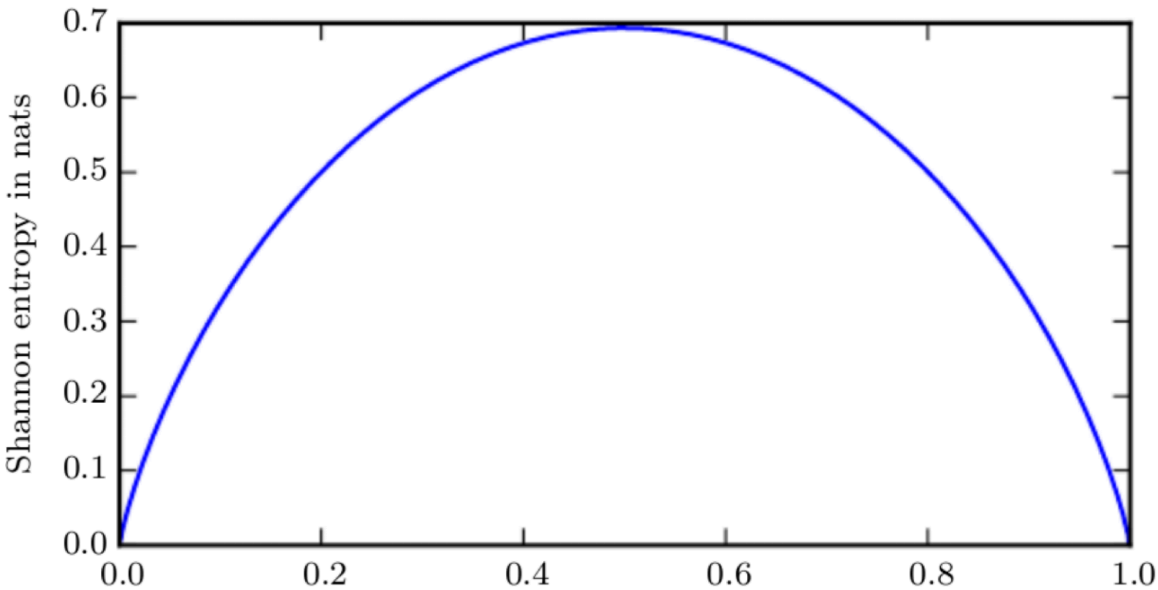


Figure 1: Example: Entropy of a biased coin with probability p

Cross-Entropy

This follows the idear behind the Cross-Entropy formular

How surprised are you on average, if you assume a distribution Q, when it's actually P.

$$H(P, Q) = \mathbb{E}_{p[h(x)]} = -\sum p(x) \log(q(x))$$

Back to loss functions

Lets start of with logistic regression, where

$$m(x) = \theta^T x = \theta_0 + \theta_1 x$$

and

$$\sigma(m(x)) = \frac{e^{\theta_0 + \theta_1 x}}{1 + e^{\theta_0 + \theta_1 x}}$$

- **Log likelihood**

$$\log(L(\theta)) = \sum_i y_i \log(\sigma(m(x:i))) + (1 - y_i) \log(1 - \sigma(m(x_i)))$$

note that flipping it gets the **Binary Cross-Entropy Loss**:

$$\mathcal{L}_{\mathrm{CE}(\theta)} = -\log(L(\theta)) = -\sum_i y_i \log(\sigma(m(x:i))) + (1 - y_i) \log(1 - \sigma(m(x_i)))$$

- **Relationship between KL Divergence and Cross Entropy**

$$D_{\mathrm{KL}(P \| Q)} = H(P, Q) - H(P)$$

$$D_{\mathrm{KL}(P \| Q)} =$$

Softmax loss

For multi-class classification with K classes, assume targets follow a Categorical distribution

then softmax

$$\sigma(m(x))_k = \frac{e^{m(x)_k}}{\sum_{j=1}^K} e^{m(x)_j}$$

The likelihood of a single observation with one-hot label $y_i \in \{0, 1\}^K$

$$p(y_i | x_i, \theta) = \prod_{k=1}^K \sigma(m(x_i))_k^{y_{ik}}$$

Now log-likelihood over all observations:

$$\log(L(\theta)) = \sum_i \sum_k y_{ik} \log(\sigma(m(x_i)))_k$$

Negating it then gives cross-entropy loss:

$\mathcal{L}_{\mathrm{CE}(\theta)} = -\log(L(\theta)) = -\sum_i \sum_k y_{ik} \log(\sigma(m(x_i)))_k$

Mean-squared error loss

Assume targets are Gaussian: $y_i \sim \mathcal{N}(m(x_i), \sigma^2)$

$$p(y_i|x_i,\theta) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(y_i - m(x_i))^2}{2\sigma^2}\right)$$

Again log likelihood over all observations:

$$\log(L(\theta)) = -\frac{n}{2}\log(2\pi\sigma^2) - \frac{1}{2\sigma^2}\sum_i (y_i - m(x_i))^2$$

Maximizing log likelihood is equivalent to minimizing

$$\mathcal{L}_{\text{MSE}} = \sum_i (y_i - m(x_i))^2$$

(because constants can be ignored for minimum)

Deep Learning intuition

Shortcut-learning

Deep neural networks tend to find the simplest possible solution, even if it is a shortcut. Shortcut-learning can break models when models are not trained on misleading data.

Figure 2: Left image is classified as a person running on a beach, because the training data had no images of cows on beaches. The right image was classified as an elephant. The images is a cat shape in elephant skin-like texture, showing that, for this model, it has some texture-bias.

The Lottery Ticket Hypothesis

Risidual connections

Residual Networks Behave Like Ensembles of Relatively Shallow Networks when unfolded

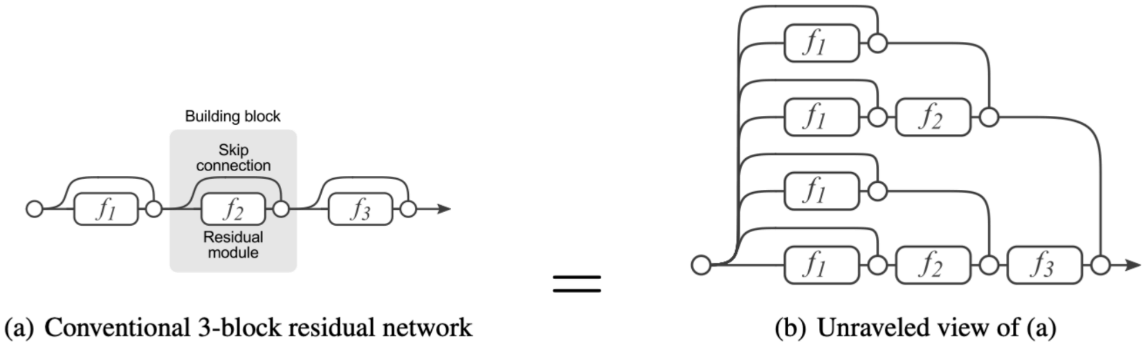


Figure 3:

Deep Double Descent

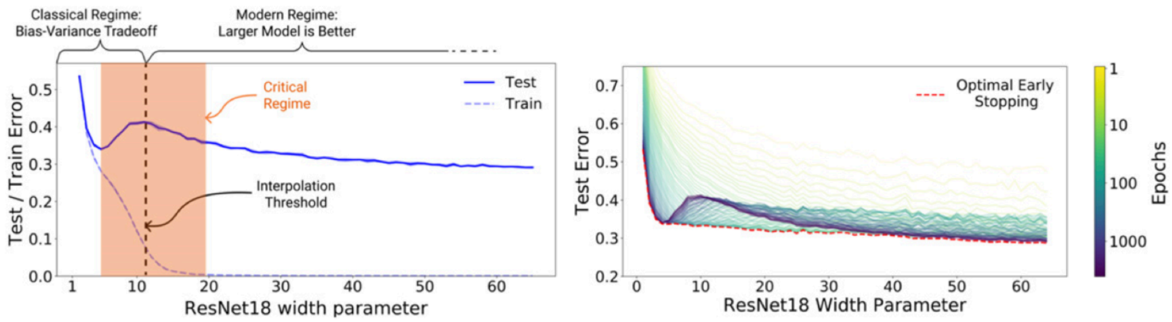


Figure 4:

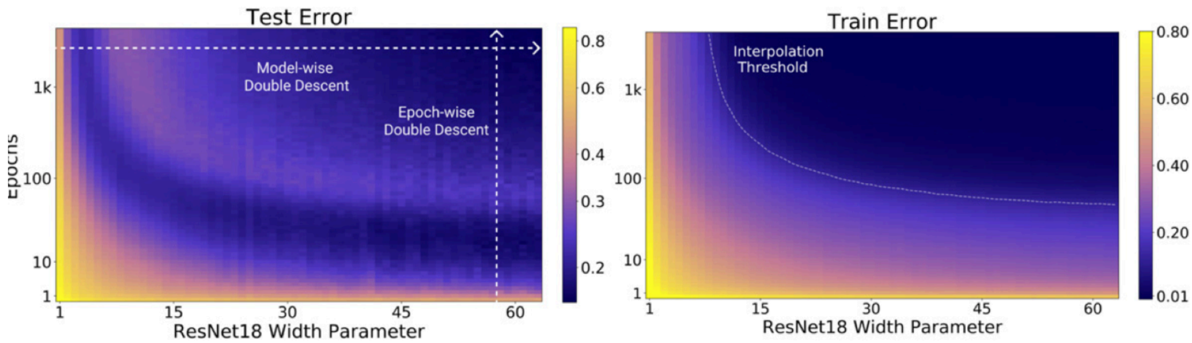


Figure 5:

Grokking

Grokking: a model first memorizes the training data, then — long after — suddenly learns to generalize

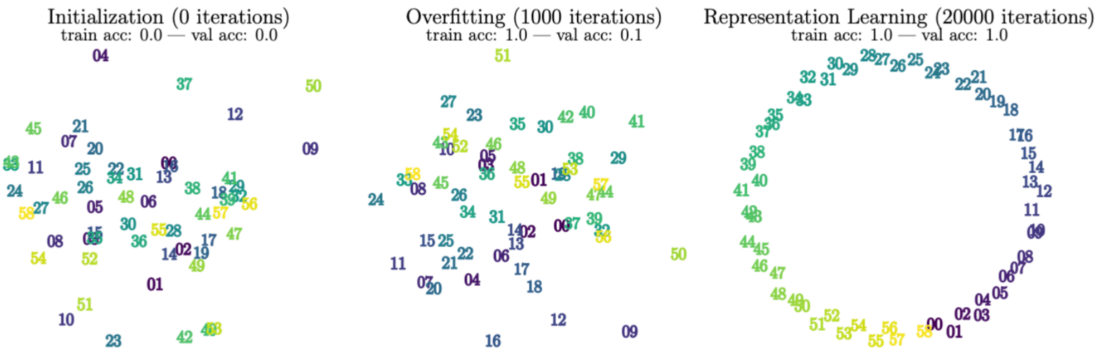


Figure 6:

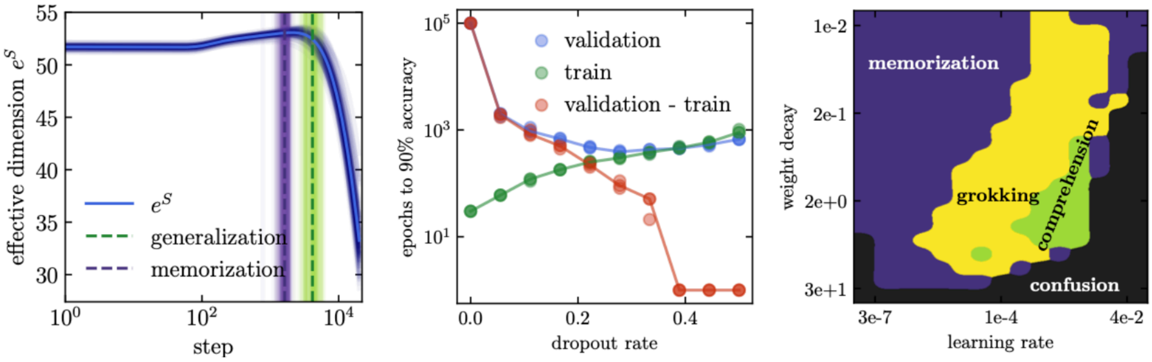
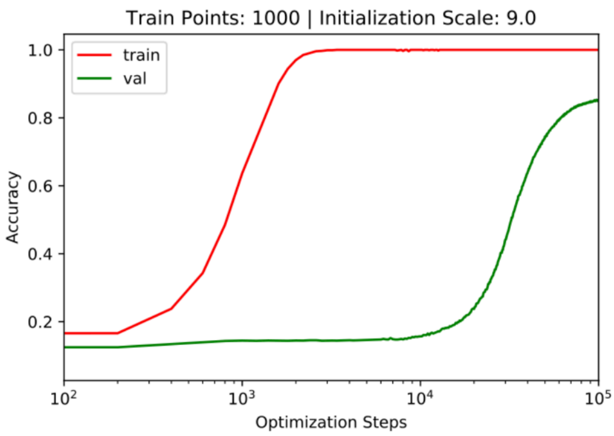
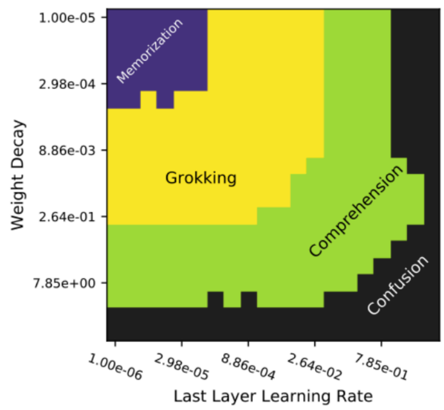


Figure 7:



(a)



(b)

Figure 8: